

# Krishna Tahiliani

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## EDUCATION

### Sardar Vallabhbhai National Institute of Technology, Surat

Surat, India

*Bachelor of Technology in Computer Science*

*Aug. 2024 – Aug. 2028*

*— CGPA: 9.46*

## EXPERIENCE

### MNNIT Allahabad

May 2026 – July 2026

*Research Intern – Deep Learning and Computer Vision*

*Remote*

- Developing a high-density person re-identification and tracking system for Kumbh Mela-scale crowd monitoring using RT-DETR, OSNet, and ByteTrack.
- Engineered temporal identity persistence across severe occlusions by combining appearance embeddings with multi-object tracking to maintain consistent person identities in dense crowds.

### Fastrac Indonesia

May 2025 – July 2025

*Software Engineering Intern – Logistics Tech*

*Remote*

- Built 10+ production-ready SEO tools including a COD vs Marketplace Estimator, PPh/PPN Calculator, and Invoice Generator, enhancing merchant decision-making and operational workflows.
- Developing a full-stack Shopify App for managing merchants, customers, and shipping rate calculation using React (TSX), GraphQL, and Prisma.
- Integrated real-time shipping logic (e.g., COD handling, dimension-based pricing) using Fastrac APIs, explored via Postman.
- Utilized Cloudflare Tunnel for secure webhook testing and local development exposure during Shopify integration.

## PROJECTS

### CrisisSync | *Next.js, PWA, Supabase, Gemini AI, Twilio, Dexie*

March – April 2026

[Github](#)

- Built an offline-first emergency reporting PWA using IndexedDB and Service Workers, ensuring reliable sync and duplicate-free submissions during connectivity loss.
- Designed a multi-channel alert pipeline triggering automated 112 calls and government dashboard notifications with exponential backoff retry (3 attempts, 1s/2s/4s) and graceful fallback on telephony failure.
- Unified emergency verification into a single Gemini multimodal API workflow, reducing latency while enforcing structured JSON validation with Zod.
- Implemented dual-timestamp validation to preserve report integrity and detect potential client-side clock manipulation.

### TransitFlow | *Node.js, React, MongoDB, Socket.IO, Google Maps, Razorpay*

May – July 2025

[GitHub](#)

- Built an urban mobility platform with carpooling, smart parking, and BRTS modules; eliminated race-condition overbooking via atomic MongoDB updates.
- Implemented Socket.IO WebSockets for real-time parking availability and emergency alerts, eliminating polling-based database load.
- Integrated Razorpay payments with HMAC-SHA256 verification and replay-attack protection across platform transactions.
- Reduced Google Places API costs by ~70% using debouncing, session-token bundling, and optimized API requests.

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, JavaScript, TypeScript, SQL

**Web & Backend:** React, Next.js, Node.js, Express.js, REST APIs, WebSockets

**Databases & Cloud:** MongoDB, Firebase

**Developer Tools:** Git, GitHub, Google Colab, Jupyter Notebook, Postman

**Core CS Foundations:** Data Structures & Algorithms, DBMS, Operating Systems, OOP, Computer Architecture

## COMPETITIVE PROGRAMMING (C++)

**LeetCode:** 1722 Rating (600+ Problems Solved)

**Codeforces:** 1311 Rating

**CodeChef:** 1490 Rating